Chloride Shift

1. In RBCs CO_2 reacts with water forming carbonic acid in a reaction that is catalysed by carbonic anhydrase.
2. The formed carbonic acid is buffered by phosphate buffer as well as haemoglobin buffer.
3. Bicarbonate returns to plasma and gets exchanged with chloride ion that enters into (shifts into) the cell if the tension of CO_2 increases in the blood.
4. On the contrary, if the CO_2 tension is reduced, chloride will exit the cell and enters the plasma.
5. In general, red blood corpuscles are impermeable to sodium or potassium whereas permeable to hydrogen, bicarbonate and chloride ions. Potassium which is a cation is available to the plasma by anionic (chloride) exchange. This results in added CO_2 being carried by plasma as sodium bicarbonate.
6. The cycle continues as CO_2 enters and passes to the red blood cells and forms carbonic acid (partial amount of which returns to plasma) by carbonic anhydrase. The remaining carbonic acid then reacts with hemoglobin buffers yielding bicarbonate that travels to the plasma in exchange of chloride and is transported.
7. The reactions mentioned so far are reversible. In lung tissue, chloride shifts back into plasma when blood becomes arterial. This eventually releases intracellular potassium to buffer the oxy-hemoglobin. In plasma, it neutralizes sodium (Figure 9).

1.6.2. Regulation by respiratory mechanism

We have seen that the carbonic acid will dissociate into CO_2 and H_2O. If there is more of H^+ within the blood more of CO_2 elimination will be carried out by lungs. If there is more of HCO_3^-, the lungs will ensure low respiratory rate thus enhancing the retention of CO_2 so that it can be useful for forming carbonic acid that can buffer the excess of bicarbonate.

The alveolei and bronchioles of lungs do perform such functions effectively. A phospholipid molecule is secreted by a particular type of cell in the lung lining the alveoli and bronchioles. This surfactant regulates (lowers) the surface tension of the alveolar membranes thereby protecting the alveoli during exhalation and inhalation.

1.6.3. Regulation by renal mechanism

The major functions of kidneys are regulation of water and electrolyte balance. This is done by excretion of waste substances in urine. The formation of urine involves three stages (i) filtration, (ii) reabsorption, (iii) secretion. The regulatory role of renal organ is achieved by the buffering capacity of the organ. For this function membrane performs key role. Passive and active transports are involved in addition to osmosis and pinocytosis. (Figure 10).

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